Distinct linear factors – 3.4

$$\frac{-x^2+x-2}{x^3-x^2+x-1} = \frac{1}{x^2+1} - \frac{1}{x-1}$$

Goal: take a rational function and rewrite it as a sum of simple pieces:

$$\frac{1}{\text{linear}}, \quad \frac{1}{\text{quadratic}}, \quad \frac{x}{\text{quadratic}}$$

Step 1 is to factor the denominator and make a guess about what simple pieces we should look for.

Suppose $\frac{p(x)}{q(x)}$ has $q(x) = (x+a_1)(x+a_2)\cdots(x+a_n)$

with $a_i \neq a_j$ for $i \neq j$.

Then try to find a decomposition of

the form $$\frac{p(x)}{q(x)} = \frac{A_1}{x+a_1} + \frac{A_2}{x+a_2} + \cdots + \frac{A_n}{x+a_n}$$

<u>Example</u>: Find $\displaystyle\int \frac{x^2+1}{x^3-x}\,dx$

$x^3 - x = x(x^2-1) = x(x+1)(x-1)$ $\quad a_1 = 0,\ a_2 = 1,\ a_3 = -1$

Goal is to find A, B, C such that

$$\frac{x^2+1}{x^3-x} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x-1}$$

$$\frac{x^2+1}{x^3-x} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x-1} \iff$$

$$x^2+1 = A(x+1)(x-1) + B(x)(x-1) + C(x)(x+1)$$

Equating coefficients

$$x^2+1 = A(x^2-1) + B(x^2-x) + C(x^2+x)$$

$$= \underline{(A+B+C)}x^2 + (-B+C)x - A$$

Coefficients must agree

$$A+B+C = 1$$

$$-B+C = 0$$

$$-A = 1 \implies A = -1$$

$$\begin{array}{r} B+C = 2 \\ -B+C = 0 \\ \hline 2C = 2 \end{array} \qquad \begin{array}{l} C = 1 \\ B = 1 \end{array}$$

substitutions

Plugging $x=0$: $1 = A(1)(-1) = -A$

$A = -1$

$x=1$: $2 = C\cdot 2 \implies C = 1$

$x=-1$: $2 = B(-1)(-2) = 2B$

$\implies B = 1$

$\boxed{A=-1,\ B=C=1 \text{ from either method}}$

$$\frac{x^2+1}{x^3-x} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x-1} = \frac{-1}{x} + \frac{1}{x+1} + \frac{1}{x-1}$$

$$\int \frac{x^2+1}{x^3-x}\,dx = \int \frac{-1}{x}\,dx + \int \frac{1}{x+1}\,dx + \int \frac{1}{x-1}\,dx$$

$$= -\ln|x| + \ln|x+1| + \ln|x-1| + C$$

__Example__ : $\displaystyle\int \frac{x^2+3x+1}{x^2-4}\,dx$

Before we look for partial fraction decomposition

$\deg(\text{num}) = 2 \geq 2 = \deg(\text{denom.}) \implies$ we should first do polynomial division!

$$\begin{array}{r@{}l} & \;\;1 \\ x^2-4 \,\big) & \overline{\;x^2+3x+1\;} \\ & -(x^2-4) \\ \hline & \;\;3x+5 \end{array}$$

$$\frac{x^2+3x+1}{x^2-4} = 1 + \underbrace{\frac{3x+5}{x^2-4}}_{\text{PFD}}$$

$$\frac{3x+5}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2} \iff 3x+5 = A(x+2) + B(x-2)$$

$x = -2$: $\quad -1 = B \cdot -4 \implies B = \frac{1}{4}$

$x = 2$: $\quad 11 = A \cdot 4 \implies A = \frac{11}{4}$

$$\int \frac{x^2+3x+1}{x^2-4}\,dx = \int 1\,dx + \int \frac{3x+5}{x^2-4}\,dx$$

$$= \int dx + \frac{11}{4}\int \frac{1}{x-2}\,dx + \frac{1}{4}\int \frac{1}{x+2}\,dx$$

$$= x + \frac{11}{4}\ln|x-2| + \frac{1}{4}\ln|x+2| + C$$